

YAYLYU, Russian Federation

WMO: 360640

Lat: 51.7715N

Lon: 87.6044E

Elev: 1576

StdP: 13.88

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.7	-5.2	-18.0	2.2	-10.4	-12.3	3.0	-4.6	16.3	25.5	15.0	25.0	5.4	320	0.555

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	81.7	64.2	78.1	63.1	74.7	61.8	66.4	77.9	64.7	74.6	63.3	71.9	5.3	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.7	90.2	69.6	61.2	85.6	67.9	59.9	81.6	66.7	32.0	77.8	30.7	74.5	29.6	72.2	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.5	11.6	9.8	-14.1	89.1	7.2	2.4	-19.3	90.8	-23.5	92.2	-27.6	93.6	-32.8	95.3		
(5)			-14.5	70.6	7.1	1.9	-19.6	71.9	-23.8	73.0	-27.8	74.1	-32.9	75.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.3	15.7	17.1	30.6	42.5	48.9	60.6	63.8	61.5	51.6	40.8	27.7	20.7		
	DBStd	18.70	10.11	10.09	9.17	8.44	7.38	6.51	5.00	5.13	6.94	7.78	9.86	9.79		
	HDD50	4887	1062	921	601	246	112	4	0	1	63	299	669	909		
	HDD65	9129	1527	1341	1066	676	499	159	83	130	404	751	1119	1374		
	CDD50	1332	0	0	0	21	79	324	427	358	110	13	0	0		
	CDD65	97	0	0	0	0	1	28	44	23	1	0	0	0		
	CDH74	861	0	0	0	0	19	340	342	148	12	0	0	0		
	CDH80	165	0	0	0	0	1	76	69	18	1	0	0	0		
	WSAvg	4.1	5.7	4.4	4.1	3.5	3.4	3.0	3.1	3.3	3.7	4.1	4.9	6.0		
	PrecAvg	33.3	0.7	0.7	1.1	3.1	4.2	4.3	5.2	4.5	3.5	2.7	2.0	1.2		
(15) Precipitation	PrecMax	43.1	1.5	1.6	3.2	6.5	8.9	7.8	8.8	9.4	6.4	7.7	4.1	2.6		
	PrecMin	24.4	0.2	0.1	0.2	1.3	1.9	2.2	2.6	2.3	1.2	1.0	0.6	0.4		
	PrecStd	4.1	0.4	0.4	0.7	1.5	1.8	1.4	1.6	1.7	1.6	1.7	1.0	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.8	42.0	55.9	70.3	76.9	86.2	85.8	82.7	75.6	62.9	53.6	42.3		
		MCWB	31.3	34.8	42.1	50.8	56.9	65.3	67.3	65.7	59.9	49.4	43.4	33.3		
	2%	DB	34.4	37.1	50.9	64.9	71.0	81.9	81.6	77.9	70.5	58.7	46.9	38.9		
		MCWB	28.7	31.2	39.3	48.4	53.7	63.1	65.3	63.5	57.3	47.4	39.0	31.8		
	5%	DB	31.3	33.9	47.3	60.3	66.4	78.0	77.9	74.5	66.7	55.7	43.2	35.3		
		MCWB	26.0	28.8	37.3	46.2	51.8	62.2	64.1	62.2	54.9	45.5	36.3	30.3		
	10%	DB	28.7	30.8	43.5	55.7	62.0	73.7	74.3	71.6	63.0	52.1	39.9	32.5		
		MCWB	24.4	25.8	35.1	44.4	50.0	60.7	63.2	60.8	52.9	43.6	35.0	27.8		
	0.4%	WB	32.3	35.3	43.3	52.8	59.0	67.2	69.7	67.2	61.5	51.3	44.1	35.1		
		MCDB	37.0	41.1	54.8	67.3	71.4	81.3	81.6	78.9	72.7	59.1	52.3	39.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.9	32.4	40.2	49.9	55.5	65.2	67.1	64.9	58.4	48.7	39.6	32.5		
		MCDB	34.1	36.5	49.4	61.8	66.4	77.9	77.9	75.1	68.6	56.7	44.6	37.6		
	5%	WB	26.5	28.8	37.8	47.4	53.5	63.5	65.4	63.5	56.2	46.5	37.3	30.6		
		MCDB	30.7	33.6	45.6	57.2	62.6	74.1	74.9	72.1	64.5	54.3	41.8	35.7		
	10%	WB	24.3	26.0	35.7	45.4	51.4	61.9	63.8	61.9	54.1	44.3	35.1	28.0		
		MCDB	28.6	30.4	42.3	53.4	60.0	71.2	72.0	69.2	60.9	51.2	39.6	32.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	15.0	16.6	18.3	19.2	20.9	17.6	16.9	16.6	13.0	10.8	10.4		
		MCDBR	11.3	14.9	20.3	27.5	28.0	29.7	26.0	23.0	22.2	18.0	12.8	10.8		
		MCWBR	7.6	10.2	11.6	13.9	14.2	13.3	11.7	10.8	10.9	9.1	7.8	6.9		
		MCDBR	10.7	14.7	18.5	22.6	22.9	24.7	21.3	19.4	19.6	16.6	12.5	11.2		
	5% WB	MCWBR	7.5	10.4	11.1	12.5	12.6	12.0	10.6	9.9	10.4	9.1	8.2	7.7		
(40) Clear-Sky Solar Irradiance	taub		0.292	0.331	0.389	0.447	0.431	0.421	0.432	0.417	0.376	0.367	0.331	0.296		
	taud		2.054	1.990	1.924	1.970	2.129	2.272	2.267	2.273	2.317	2.197	2.075	2.016		
	Ebn at Noon		214	238	246	248	261	265	259	257	254	226	194	185		
	Edh at Noon		25	36	48	52	46	41	40	38	32	30	25	23		
(44) All-Sky Solar Radiation	RadAvg		403	736	1175	1432	1649	1852	1716	1480	1107	626	356	283		
	RadStd		40	49	102	109	168	167	101	109	89	74	37	33		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page